

Institute of Actuaries of India

Subject CT6 – Statistical Methods

October 2014 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:-

(i) Using the formula for $f(x)$ and the substitution $t = (\log x - \mu)/\sigma - r\sigma$ gives the reqd.

$$\begin{aligned} \text{Probability} &= \int_A^B f(x) dx = \int_A^B \frac{1}{x\sigma\sqrt{2\pi}} \exp(-0.5((\log x - \mu)/\sigma - r\sigma)^2) dx \\ &= \int_{A_r}^{B_r} \frac{1}{\sqrt{2\pi}} \exp(-0.5(t + r\sigma)^2) dt \\ &= \exp(r\mu + 0.5 r^2 \sigma^2) [\Phi(B_r) - \Phi(A_r)]. \end{aligned}$$

[6]

(ii) In the above formula putting $r = 1$, $A = 0$ & $B = \infty$ and using $\Phi(-\infty) = 0$, $\Phi(0) = 0.5$ & $\Phi(\infty) = 1$,

$$\text{Mean} = E(X) = \exp(\mu + 0.5 \sigma^2) [\Phi(\infty) - \Phi(-\infty)] = \exp(\mu + 0.5 \sigma^2).$$

Putting $r = 2$ & same values of A & B

$$E(X^2) = \exp(2\mu + 2 \sigma^2) [\Phi(\infty) - \Phi(-\infty)] = \exp(2\mu + 2 \sigma^2)$$

$$\text{So, Variance} = E(X^2) - (E(X))^2 = \exp(2\mu + 2 \sigma^2) - \exp(2\mu + \sigma^2) = \exp(2\mu + \sigma^2)(\exp(\sigma^2) - 1).$$

Putting $r = k$ & same values of A & B , The k th order non central moment is

$$E(X^k) = \exp(k\mu + k^2 \sigma^2 / 2) [\Phi(\infty) - \Phi(-\infty)] = \exp(k\mu + k^2 \sigma^2 / 2).$$

[4]

(iii) With this reinsurance treaty the insurer pays first 25000 (M) of each Claim X . So the net amount paid by the insurer (Y) is given as $Y = X$ for $X \leq M$ & $Y = 25000$ for $X > M$.

$E(Y) = \int_0^M x f(x) dx + M \int_M^\infty f(x) dx$, the first integral is from 0 to M & the 2nd is from M to ∞ .

$$M_0 = (\log 25000 - 8.5) / 0.8 = 2.033. \text{ (Putting } r = 0 \text{ in the above formula)}$$

$$M_1 = (\log 25000 - 8.5) / 0.8 - 0.8 = 1.233. \text{ (Putting } r = 1 \text{ in the above formula)}$$

$$\text{Thus the mean } E(Y) = \exp(\mu + 0.5 \sigma^2) [\Phi(M_1) - \Phi(-\infty)] + 25000 [\Phi(\infty) - \Phi(M_0)]$$

$$= 6768(\Phi(1.233) + 25000(1 - \Phi(2.033)) = 6557.$$

$$E(Y^2) = \int_0^M x^2 f(x) dx + M^2 \int_M^\infty f(x) dx, \text{ the first integral is from 0 to } M \text{ & the 2}^{nd} \text{ is from } M \text{ to } \infty.$$

$$\text{As above } M_0 = (\log 25000 - 8.5) / 0.8 = 2.033. \text{ (Putting } r = 0 \text{ in the above formula)}$$

$$M_2 = (\log 25000 - 8.5) / 0.8 - 2 * 0.8 = 0.433. \text{ (Putting } r = 2 \text{ in the above formula)}$$

$$\text{So, } E(Y^2) = \exp(2\mu + 2 \sigma^2) [\Phi(M_2) - \Phi(-\infty)] + (25000^2) [\Phi(\infty) - \Phi(M_0)]$$

$$= 86876663 [\Phi(0.433)] + (25000^2) [1 - \Phi(2.033)] = 71135800.$$

$$\text{Hence } \text{Var}(Y) = E(Y^2) - E(Y)^2 = 71135800 - 6557^2 = 5304^2.$$

[8]

[18 Marks]

Solution 2:-

$$\begin{aligned}
 \text{(i)} \quad X_t &= \alpha X_{t-1} + Z_t = \alpha(\alpha X_{t-2} + Z_{t-1}) + Z_t = \alpha^2 X_{t-2} + \alpha Z_{t-1} + Z_t \\
 &= \alpha^2 (\alpha X_{t-3} + Z_{t-2}) + \alpha Z_{t-1} + Z_t \\
 &= \alpha^3 X_{t-3} + Z_t + \alpha Z_{t-1} + \alpha^2 Z_{t-2}
 \end{aligned}$$

Hence continuing as above $X_t = \sum C_j Z_{t-j}$, where C_j 's are constant and $C_j = \alpha^j$ and the summation ranges from 0 to ∞ . [2]

$$\begin{aligned}
 \text{(ii)} \quad E(X_t) &= E(\sum C_j Z_{t-j}) = \sum C_j E(Z_{t-j}) = 0. \\
 \text{Var}(X_t) &= \text{Var}(\sum C_j Z_{t-j}) = \sum \alpha^{2j} \text{Var}(Z_{t-j}) = \sum \alpha^{2j} \sigma^2 = \sigma^2 / (1 - \alpha^2). \quad (\text{Since } |\alpha^2| < 1).
 \end{aligned}$$

[2]

(iii) X_t is the sum of white noise terms which are normal random variables. Since the sum of normal random variables is itself a normal random variable, we have that X_t is a normal random variable with mean 0 and variance $\sigma^2 / (1 - \alpha^2)$. (as calculated above)

When $t = 10$, $\alpha = 0.6$ and $\sigma^2 = 50$, we have:

X_{10} is a normal random variable with mean 0 & variance 78.125.

So, $P(X_{10} > 10) = P(Z > 10/\sqrt{78.125}) = P(Z > 1.13137) = 1 - 0.87105 = 0.12895$, where Z is a standard normal variate.

[3]

(iv) For $\alpha > 1$ the mean would be unchanged, but the variance would be huge (as the infinite sum would no longer be convergent). Therefore the probability of being greater than 10 would be $P(Z > 0) = 0.5$.

[2]

[9 Marks]

Solution 3:-

$P(N - r = k | r \text{ big claims}) = P(N = r + k) * P(k \text{ small claims among } r + k \text{ claims}) / P(r \text{ big claims})$

$$\begin{aligned}
 P(r \text{ big claims}) &= \sum_{j=0}^{\infty} e^{-\lambda} \frac{\lambda^{r+j}}{(r+j)!} * \frac{(r+j)!}{r!j!} p^r (1-p)^j \\
 &= \frac{\lambda^r}{r!} p^r * \sum_{j=0}^{\infty} e^{-\lambda} \frac{\lambda^j}{j!} (1-p)^j \\
 &= \frac{\lambda^r}{r!} p^r e^{-\lambda} e^{\lambda(1-p)}
 \end{aligned}$$

Thus, substituting values of each probability, we get the following:

$$\begin{aligned}
 P(N - r = k | r \text{ big claims}) &= \frac{e^{-\lambda} \frac{\lambda^{r+k}}{(r+k)!} \frac{(r+k)!}{r!k!} p^r (1-p)^k}{\frac{\lambda^r}{r!} p^r e^{-\lambda} e^{\lambda(1-p)}} \\
 &= \frac{\lambda^k (1-p)^k}{k!} e^{-\lambda(1-p)}
 \end{aligned}$$

This is the probability from a poisson distribution with parameter $\lambda(1-p)$. Hence conditional mean of $N - r$ is $\lambda(1-p)$.

Solution 4:-

(i) Let N denote the number of accidents. Then $N \sim B(500, p)$.

$$M_N(t) = (p e^t + 1 - p)^{500}$$

If an accident happens, total cost of replacement tyres, X, has the following distribution:

(No. of tyres)	N	0	1	2	3	4
(Probability)	p	0.6561	0.2916	0.0486	0.0036	0.0001
(Cosy in units)	X	0	5	10	15	20

$$M_X(t) = 0.6561 + 0.2916 e^{5t} + 0.0486 e^{10t} + 0.0036 e^{15t} + 0.0001 e^{20t}$$

$$M_S(t) = M_N(\log M_X(t))$$

$$= (p e^{\log M_X(t)} + p - 1)^{500}$$

$$= (p M_X(t) + 1 - p)^{500}$$

$$= (p(0.6561 + 0.2916 e^{5t} + 0.0486 e^{10t} + 0.0036 e^{15t} + 0.0001 e^{20t}) + 1 - p)^{500}$$

[5]

(ii) $E(N) = 500 p$

$$V(N) = 500 p (1-p)$$

$$E(X) = 0.6561 * 0 + 0.2916 * 5 + 0.0486 * 10 + 0.0036 * 15 + 0.0001 * 20 = 2$$

$$V(X) = E(X^2) - E(X)^2 = 0.6561 * 0 + 0.2916 * 25 + 0.0486 * 100 + 0.0036 * 225 + 0.0001 * 400 - 2^2$$

$$= 13 - 4 = 9$$

$$E(S) = E(N) E(X)$$

$$E(S) = 500 p * 2 = 1000 p$$

$$V(S) = E(N) V(X) + V(N) (E(X))^2$$

$$= 500 p * 9 + 500 p (1-p) * 4$$

$$= 4500 p + 2000 p - 2000 p^2$$

$$= 6500 p - 2000 p^2$$

[6]

(iii) Let R denote the number of tyres needing replacement. Then $R \sim B(2000, 0.1 p)$ and $T = 5 R$

Thus $E(T) = 5 E(R) = 5 * 2000 * 0.1 p = 1000 p$

$$\begin{aligned} \text{And Var}(T) &= \text{Var}(5R) = 25 V(R) = 25 * 2000 * 0.1 p * (1 - 0.1 p) \\ &= 5000 p (1 - 0.1 p) \end{aligned}$$

[2]

(iv) (a) If $p = 0.05$ then

$$E(S) = E(T) = 1000 * 0.05 = 50$$

$$\begin{aligned} V(S) &= 6500 * 0.05 - 2000 * 0.05^2 \\ &= 320 \end{aligned}$$

$$\begin{aligned} V(T) &= 5000 * 0.05 * (1 - 0.1 * 0.05) \\ &= 248.75 \end{aligned}$$

(b) Assuming both S & T can be approximated by normal distribution, and allowing for continuity correction:

$$P(S > 77.5) \approx P(N(0,1) > \frac{77.5-50}{17.88}) = P(N(0,1) > 1.5380) = 0.06202$$

$$P(T > 77.5) \approx P(N(0,1) > \frac{77.5-50}{15.77}) = P(N(0,1) > 1.7438) = 0.0406$$

(c) The two distributions have same mean but different variances, it being higher for S compared to T. This means the probability would be higher under S than under T. Though the probability under the two distribution is small in absolute terms, it's still higher by 50% under S compared to T.

[5]

[18 Marks]

Solution 5:-

(i) (a) The prior density for λ is $f(\lambda) = \frac{e^{-\lambda} \lambda^\mu}{\Gamma(\mu)}$ and likelihood function for $x_1, x_2, \dots, x_n$ is:

$$f(x|\lambda) = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

Thus the posterior density is

$$f(\lambda|x) \propto f(\lambda) * f(x|\lambda)$$

$$\propto e^{-\frac{\lambda}{\mu}} * e^{-n\lambda} * \lambda^{\sum_{i=1}^n x_i}$$

$$= \lambda^{\sum x_i} \exp(-\lambda (n + \frac{1}{\mu}));$$

This pdf resembles Gamma distribution: Gamma ($1 + \sum_{i=1}^n x_i$, $n + \frac{1}{\mu}$)

(b) The Bayesian estimate under quadratic loss is the posterior mean:

$$\frac{\sum_{i=1}^n x_i + 1}{n + \frac{1}{\mu}} = \frac{n}{n + \frac{1}{\mu}} \bar{x} + \frac{\frac{1}{\mu}}{n + \frac{1}{\mu}} \mu$$

Thus the credibility factor is $\frac{n}{n + \frac{1}{\mu}}$.

(c) With the given values, the estimate of λ is $\frac{650+1}{(10+\frac{1}{50})} = 64.97$ [6]

(ii)

(a) If any variable follows Pareto (α , v), then

$$\text{Mean} = \frac{v}{\alpha-1} = 2 \text{ and Variance} = \frac{\alpha v^2}{(\alpha-2)(\alpha-1)^2} = 12$$

Solving these two equations gives $\alpha = 3$ and $v = 4$.

Let X denote a randomly chosen call, then

$$f(x) = 0.6 e^{-x} + 0.4 \frac{3 \cdot 4^3}{(4+x)^4}$$

Then

$$P(X > M) = 0.6 e^{-M} + 0.4 \left(\frac{4}{4+M} \right)^3$$

(b)

Let T be the total time spent answering all calls from one day:

$$E(T) = \lambda (0.6 * 1 + 0.4 * 2) = 64.97 * 1.4 = 90.96$$

[2]

(c)

Let R be the time to answer a call under the new strategy, thus $R = \min(X, 5)$

$$E(R) = E(\min(X, 5)) = \int_0^5 x f(x) dx + 5 P(X > 5)$$

$$= E(X) - \int_0^\infty x f(x+5) dx$$

The reduction in in expectation per call is:

$$\begin{aligned}
 \int_0^{\infty} x f(x+5) dx &= \int_0^{\infty} x \left(0.6 e^{-x-5} + 0.4 \frac{3 * 4^3}{(4+5+x)^4} \right) dx \\
 &= 0.6 e^{-5} \int_0^{\infty} x e^{-x} dx + \frac{0.4 * 4^3}{9^3} \int_0^{\infty} x \frac{3 * 9^3}{(9+x)^4} dx \\
 &= 0.6 e^{-5} + 0.4 * \left(\frac{4}{9} \right)^3 * \frac{9}{2} \\
 &= 0.1620
 \end{aligned}$$

Hence the total reduction in one day's call is $90.96 * 0.1620 = 14.74$ minutes.

[10]

[16 Marks]

Solution 6:-

(i) Let N_i denote the number of cargo ships covered in year i . Let $N = \sum N_i = 1049$

Let M_i denote the number of claims from cargo ships in year i . Let $M = \sum M_i$

Then we know that $M \sim P(N\lambda)$. The likelihood is given by:

$$L = C e^{-1049\lambda} \frac{(1049\lambda)^m}{m!}$$

From the data, $m = 310$ over the last 5 years. The log likelihood is given by:

$$L = \log L = D - 1049\lambda + m \log(1049\lambda)$$

Differentiating this

$$\frac{dl}{d\lambda} = -1049 + \frac{m}{\lambda}, \text{ setting this equal to 0, gives}$$

$$\Rightarrow \hat{\lambda} = \frac{m}{1049} = \frac{310}{1049} = 0.2955$$

Double differentiation being negative confirms that this is the maximum.

[5]

(ii) We first need to calculate

$$\begin{aligned}
 P^* &= \frac{1}{5 * 3 - 1} \sum_{i=1}^3 \bar{P}_i \left(1 - \frac{\bar{P}_i}{\bar{P}} \right) \\
 &= \frac{1}{14} \left[1049 \left(1 - \frac{1049}{3807} \right) + 2282 \left(1 - \frac{2282}{3807} \right) + 476 \left(1 - \frac{476}{3807} \right) \right] \\
 &= 149.33
 \end{aligned}$$

The estimators are as follows:

$$E(m(\theta)) = \bar{X} = 0.264775$$

$$E(s^2(\theta)) = \frac{1}{3} \sum_{i=1}^3 \frac{1}{5-1} \sum_{j=1}^5 P_{ij} \left(\frac{Y_{ij}}{P_{ij}} - \bar{X}_i \right)^2$$

$$= \frac{1}{12} * (1.7192 + 2.42 + 0.7846) = 0.4103$$

$$Var(m(\theta)) = \frac{1}{P^*} \left(\frac{1}{3 * 5 - 1} \sum_{i=1}^3 \sum_{j=1}^5 P_{ij} \left(\frac{Y_{ij}}{P_{ij}} - \bar{X} \right)^2 - E(s^2(\theta)) \right)$$

$$= \frac{1}{149.33} \left(\frac{1}{14} (2.7108 + 3.2768 + 1.0854) - .4103 \right) = 0.0006356$$

Thus the credibility factor for cargo policies is given by:

$$Z_C = \frac{P_1}{P_1 + \frac{E(s^2(\theta))}{Var(m(\theta))}} = \frac{1049}{1049 + \frac{0.4103}{0.0006356}} = 0.61905$$

Number of claims per unit risk is then given by:

$$0.61905 * 0.2955 + (1 - 0.61905) * 0.264775 = 0.2838$$

And so expected claims from cargo ships in year 6 are $400 * 0.2838 = 113.52$

[6]

(iii) The main differences are:

- The approach in (i) uses only the data from cargo ships policies; the approach in (ii) uses a weighted average of the data from cargo ship policies and the overall data.
- The approach in (i) makes a precise distributional assumption about claims (i.e. that they are Poisson distributed). This assumption is not used in approach (ii).

Any other point

[3]

[14 Marks]

Solution 7:-

i. Probability of ruin on first claim

First, assume that the first claim occurs at time t . We want the probability that the first claim is big enough to cause ruin, ie it is bigger than the surplus accumulated at time t .

Since the initial surplus is U and the rate of premium income is 1 per day, the probability that the first claim will cause ruin if it occurs at time t is:

$$\int_{(U+1 \times t)}^{\infty} \lambda e^{-\lambda x} dx = \left[\frac{-\lambda e^{-\lambda x}}{\lambda} \right]_{(U+t)}^{\infty} = e^{-\lambda(U+t)}$$

We now consider all the times at which the first claim could occur. Since we have a Poisson process, the time to the first claim will have an exponential distribution with parameter of μ .

Hence the probability of ruin at the first claim is:

$$\int_0^{\infty} \mu e^{-\mu t} e^{-\lambda(U+t)} dt = \left[\frac{\mu e^{-\lambda U}}{\mu + \lambda} \right] \int_0^{\infty} (\mu + \lambda) e^{-t(\mu + \lambda)} dt = \left[\frac{\mu e^{-\lambda U}}{\mu + \lambda} \right]$$

But since this integral is just the density function of another exponential distribution, it integrates to 1. So the unconditional probability of ruin at the first claim is:

$$\left[\frac{\mu e^{-\lambda U}}{\mu + \lambda} \right]$$

[6]

ii. **Size of surplus for probability of ruin less than 1%**

The same can be calculated by solving the following equation:

$$\left[\frac{\mu e^{-\lambda U}}{\mu + \lambda} \right] < 1\% \text{ or } 0.01$$

or

$$\left[\frac{0.01 e^{-0.0125U}}{0.01 + 0.0125} \right] < 0.01$$

Or

$$e^{-0.0125U} < 0.0225$$

Or

$$U > \ln(0.0225) / (-0.0125)$$

Or

$$U > 303.54$$

[2]

[8 Marks]

Solution 8:-

(i) (a) Generate Uniform Observations

If $Y \sim U(-1, 1)$ then $F(y) = [y - (-1)] / [1 - (-1)] = (y+1)/2$

To generate observations, y , using the inverse transforms method on a pseudo-random number u :

$$u = F(y) = (y+1)/2$$

or

$$y = 2u - 1$$

This gives:

$$-0.078, 0.932, 0.848$$

(b) Acceptance Rejection Method

The PDF of the U (-1, 1) is $h(x) = 1/2$ where $-1 < x < 1$.

The probability of accepting a value x generated from U (-1,1) is:

$$g(x) = f(x) / Ch(x)$$

$$\text{where } C = \max_{-1 \leq x \leq 1} \left\{ \frac{f(x)}{h(x)} \right\} = \max_{-1 \leq x \leq 1} \{1.2 - 0.6 x^2\}$$

Differentiating to find the maximum and setting the derivative equal to zero gives:

$$\frac{d}{dx} \{1.2 - 0.6 x^2\} = -1.2x = 0 \Rightarrow x = 0$$

$$\frac{d^2}{dx^2} \{1.2 - 0.6 x^2\} = -1.2 < 0 \Rightarrow \text{max}$$

Hence by substituting $x = 0$, we get $C = 1.2 - 0.6 \times 0^2 = 1.2$.

Therefore the probability of accepting a value x generated from U (-1, 1) is:

$$g(x) = \frac{f(x)}{Ch(x)} = \frac{0.6 - 0.3 x^2}{1.2 \times 0.5} = 1 - 0.5 x^2$$

For $x = -0.078$ the probability of accepting this value is:

$$g(-0.078) = 1 - 0.5 (-0.078)^2 = 0.9970$$

Since $0.843 < g(-0.078) = 0.9970 \rightarrow \text{accept } -0.078$.

Similarly for $x = 0.932$ we have $g(0.932) = 0.5657$.

Since $0.427 < g(0.932) = 0.5657 \rightarrow \text{accept } 0.932$

Finally for $x = 0.848$ we have $g(0.848) = 0.6404$.

Since $0.683 > g(0.848) = 0.6404 \rightarrow \text{reject } 0.848$.

So the simulated values of X are: $-0.078, 0.932$

[8]

ii) Proportion Rejected

The proportion of simulations accepted in the long run is $1 / C = 1 / 1.2 = 5/6$. Hence the proportion of simulations rejected in the long run is $1/6$.

[1]***iii) Comparison of methods***

Integrating the density function would give a cubic distribution function, $F(x)$. Inverting a cubic function to get $x = F^{-1}(u)$ is not easy to do.

[1]**[10 Marks]**
